2. Consider the linear programming problem (primal) of minimizing $c^T x$ subject to $Ax = b,\ x \geq 0$. Let x^* be an optimal solution to the primal, assumed to exist, and let π^* be an optimal solution to the dual.

a) Let $\hat{x}$ be an optimal solution to the primal, where c is replaced by some $\hat{c}$. Show that $(\hat{c} - c)^T(\hat{x} - x^*) \leq 0$.

b) Let $\tilde{x}$ be an optimal solution to the primal, where b is replaced by some $\tilde{b}$ (the cost vector fixed at c). Show that $(\pi^*)^T(\tilde{b} - b) \leq c^T(\tilde{x} - x^*)$.

2. a)
$$\begin{aligned} \min\ & c^T x \\ \text{s.t. } & Ax = b \\ & x \geq 0 \end{aligned} \qquad \begin{aligned} \min\ & \hat{c}^T x \\ \text{s.t. } & Ax = b \\ & x \geq 0 \end{aligned}$$

Since the x^* and $\hat{x}$ are the optimal solution of the two primal problem respectively,

we have :
$$c^T x^* \leq c^T \hat{x} \quad \text{and}$$
$$\hat{c}^T \hat{x} \leq \hat{c}^T x^*$$

then : $(\hat{c} - c)^T(\hat{x} - x^*) = \hat{c}^T(\hat{x} - x^*) + c^T(x^* - \hat{x}) \leq 0$

2. b) For $\begin{cases} \min\ c^T x \\ \text{s.t. } Ax = b \\ \quad x \geq 0 \end{cases}$ and its dual $\begin{cases} \max\ b^T \pi \\ \text{s.t. } A^T \pi \leq c \\ \quad \pi \text{ u.r.s.} \end{cases}$

we have : $c^T x^* = b^T \pi^*$ and $A^T \pi^* \leq c$

$\left((\pi^*)^T A \leq c^T \right)$

For $\begin{cases} \min\ c^T x \\ \text{s.t. } Ax = \tilde{b} \\ \quad x \geq 0 \end{cases}$

we have : $(\pi^*)^T \tilde{b} = (\pi^*)^T (A\tilde{x}) = ((x^*)^T A)\tilde{x} \leq c^T \tilde{x}$

So : $(\pi^*)^T(\tilde{b} - b) = (\pi^*)^T \tilde{b} - (\pi^*)^T b$

$= (\pi^*)^T \tilde{b} - c^T x^* \leq c^T \tilde{x} - c^T x^*$

$= c^T(\tilde{x} - x^*)$